Colorimetric Analysis of Chemical Test Strips: Mathematical Methods for a Bayer-Camera Colorimeter

Design Notes

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1 Introduction and Instrument Overview

This document describes the mathematical methods developed for a closed-box colorimeter designed to measure arsenic concentration in water using chemical test strips. The strips discolour in the range white – yellow – brown, and the highest sensitivity is required in the pale-yellow region where human colour discrimination is poor.

1.1 Hardware

- **Camera:** Sony IMX219, 8 MP, controlled by a Raspberry Pi Zero. Raw Bayer images are captured; the four channels are B, G_b, G_r, R.
- **Illumination:** Narrow-band LEDs (blue, green, red, and optionally more), each used alone. LED spectral output is temperature-dependent.
- **Stage:** A reference colour card with invariant patches (white, black, and several shades of yellow to brown) surrounds the test strip at the centre of the field.

1.2 Key Design Principles

1. Every image contains both the unknown test strip *and* all invariant reference patches. Distances are computed *within* a single image, making them independent of LED drive current and exposure time.
2. LED temperature is retained as a continuous covariate because it shifts the LED spectrum in a way that is not cancelled by within-image normalisation.
3. Calibration uses many test strips of known concentration spanning the measurement range. Each strip produces a vector of colour distances; the GP maps these to concentration.

2 Image Channels and Normalisation

2.1 Multi-spectral Channel Vector

Each Bayer image under one LED colour yields four channels. Only channels with sufficient signal-to-noise ratio (SNR) are used. The SNR criterion for channel k under LED i is

$$\mathrm{SNR}_{ik} = \frac{\bar{x}_{ik}^{\mathrm{white}}}{\sigma_{ik}^{\mathrm{dark}}} \geq \tau, \tag{1}$$

where $\bar{x}_{ik}^{\mathrm{white}}$ is the mean pixel value of the white reference patch and $\sigma_{ik}^{\mathrm{dark}}$ is the standard deviation of the dark (black patch) signal. The threshold τ (typically 10) is fixed for all LEDs. For example, the R channel is near zero under blue illumination and is excluded.

The set of channels passing the SNR test for LED i is denoted $\mathcal{C}_i \subseteq \{B, G_b, G_r, R\}$. The total number of included channels across all LEDs is $N = \sum_i |\mathcal{C}_i|$, and $N > 4$ in general.

2.2 Normalisation

Raw pixel means are normalised to remove illumination intensity and exposure-time variation. For channel k, patch j, LED i:

$$\tilde{x}_{ijk} = \frac{\bar{x}_{ijk}}{t_{\mathrm{exp},i}\ \bar{x}_{ik}^{\mathrm{white}}}, \tag{2}$$

where $t_{\mathrm{exp},i}$ is the exposure time for LED i and $\bar{x}_{ik}^{\mathrm{white}}$ is from the *same image*. Both strip and reference patches are divided by the same white mean, so the ratio is dimensionless and

illumination-intensity-independent. LED temperature drift is *not* removed by this normalisation and is handled explicitly as a regression covariate.

The normalised colour vector for a patch j under LED i is

$$\tilde{\mathbf{x}}_{ij} = \big(\tilde{x}_{ijk}\big)_{k\in\mathcal{C}_i} \in \mathbb{R}^{|\mathcal{C}_i|}. \tag{3}$$

3 Covariance Estimation

3.1 Pooled Within-Patch Covariance

The Mahalanobis distance requires an estimate of the within-patch pixel noise covariance $\mathbf{S}$. This is estimated once during calibration by pooling per-pixel observations from all invariant reference patches (excluding SAMPLE and BLACK) across all calibration images taken with the same LED.

Let $\mathbf{x}_{ij\ell}$ denote the normalised $|\mathcal{C}_i|$-vector of pixel ℓ in reference patch j of a calibration image under LED i, and let n_{ij} be the pixel count of that patch. The pooled sample covariance is

$$\mathbf{S}_i = \frac{1}{\Big(\sum_j n_{ij}\Big) - J} \sum_j \sum_{\ell=1}^{n_{ij}} \big(\mathbf{x}_{ij\ell} - \bar{\mathbf{x}}_{ij}\big)\big(\mathbf{x}_{ij\ell} - \bar{\mathbf{x}}_{ij}\big)^\top, \tag{4}$$

where J is the number of reference patches used. Ledoit–Wolf shrinkage is applied to stabilise the estimate:

$$\hat{\mathbf{S}}_i = (1-\lambda)\,\mathbf{S}_i + \lambda\,\hat{\sigma}^2\mathbf{I}, \qquad \hat{\sigma}^2 = \frac{1}{p}\operatorname{tr}(\mathbf{S}_i), \tag{5}$$

where $\lambda \in (0,1)$ is chosen analytically to minimise the expected squared Frobenius error. The result $\hat{\mathbf{S}}_i$ is positive definite with controllable condition number.

3.2 Bayesian Update at Measurement Time

When a new image is captured, the live reference patches provide fresh pixel noise observations. The calibration estimate is used as an inverse-Wishart prior; the live scatter matrix updates it.

Prior.

$$\mathbf{S} \sim \mathcal{W}^{-1}(\boldsymbol{\Psi}_0,\, \nu_0), \qquad \boldsymbol{\Psi}_0 = \nu_0\,\hat{\mathbf{S}}_i, \tag{6}$$

so that $\mathbb{E}[\mathbf{S}] = \hat{\mathbf{S}}_i$. The prior weight ν_0 is set to the total number of strided calibration pixels used to estimate $\hat{\mathbf{S}}_i$, reflecting the confidence accumulated during calibration.

Live scatter matrix. From the n_{live} stride-subsampled pixels of the live reference patches:

$$\mathbf{A}_{\text{live}} = \sum_{j\in\mathcal{R}} \sum_{\ell=1}^{n_j} \big(\mathbf{x}_{j\ell} - \bar{\mathbf{x}}_j\big)\big(\mathbf{x}_{j\ell} - \bar{\mathbf{x}}_j\big)^\top, \tag{7}$$

where $\mathcal{R}$ denotes the set of invariant reference patches and $n_{\text{live}} = \sum_{j\in\mathcal{R}} n_j$.

Posterior mean.

$$\mathbf{S}_{\text{updated}} = \frac{\boldsymbol{\Psi}_0 + \mathbf{A}_{\text{live}}}{\nu_0 + n_{\text{live}} - p - 1}, \tag{8}$$

where $p = |\mathcal{C}_i|$ is the number of channels. When $n_{\text{live}} \ll \nu_0$ the posterior is dominated by the calibration prior; when $n_{\text{live}} \gg \nu_0$ the live data dominates. The update is most significant when LED temperature has drifted from calibration conditions.

4 Mahalanobis Distance

4.1 Definition

The Mahalanobis distance between the normalised colour vector of the test strip $\tilde{\mathbf{x}}_{\text{strip}}$ and that of reference patch r, both in the same image under LED i, is

$$d_M(\tilde{\mathbf{x}}_{\text{strip}}, \tilde{\mathbf{x}}_r) = \sqrt{\left(\tilde{\mathbf{x}}_{\text{strip}} - \tilde{\mathbf{x}}_r\right)^\top \mathbf{S}_i^{-1} \left(\tilde{\mathbf{x}}_{\text{strip}} - \tilde{\mathbf{x}}_r\right)}. \tag{9}$$

The squared distance d_M^2 is asymptotically χ^2_p distributed under the hypothesis that the strip colour is drawn from the same distribution as reference r.

4.2 Why Within-Image Distances Are Stable

Because both $\tilde{\mathbf{x}}_{\text{strip}}$ and $\tilde{\mathbf{x}}_r$ are normalised by the same white patch mean from the same image (equation (2)), the distance is:

- independent of LED drive current and total flux;
- independent of camera exposure and gain;
- partially self-correcting for LED temperature drift (both patches shift together under a spectral change).

Residual temperature sensitivity arises from the interaction of the LED spectral shift with the differential reflectance spectra of the strip and the reference patches. This is retained as a covariate in the GP.

4.3 Outlier Detection

A distance is flagged as a statistical outlier if

$$d_M^2 > \chi^2_{p,\,1-\alpha}, \tag{10}$$

where $\chi^2_{p,\,1-\alpha}$ is the $(1-\alpha)$ quantile of the chi-squared distribution with p degrees of freedom. At $\alpha = 0.01$ and $p = 4$ channels, the threshold is approximately 13.28. A flagged distance indicates a possible contaminated strip, wrong reagent, or sensor artefact.

5 Feature Vector and Calibration Dataset

5.1 Feature Vector

For a test strip image set (one image per LED), the GP input feature vector is assembled as follows. Let I denote the number of LEDs and $\mathcal{R} = \{r_1, \ldots, r_J\}$ the set of reference patches. The feature vector is

$$\mathbf{f} = \Big(d_{11},\, d_{12},\, \ldots,\, d_{IJ},\; T_1,\, T_2,\, \ldots,\, T_I\Big) \in \mathbb{R}^{IJ+I}, \tag{11}$$

where $d_{ij} = d_M(\tilde{\mathbf{x}}_{\text{strip}},\, \tilde{\mathbf{x}}_{r_j})$ is the Mahalanobis distance to reference r_j under LED i, and T_i is the measured temperature of LED i at capture time.

5.2 Calibration Dataset

Each calibration strip (known concentration c) yields one feature vector $\mathbf{f}$ and one target c. Repeating for M strips:

$$\mathcal{D} = \big\{(\mathbf{f}^{(m)},\, c^{(m)})\big\}_{m=1}^{M}, \tag{12}$$

with $\mathbf{f}^{(m)} \in \mathbb{R}^{IJ+I}$ and $c^{(m)} \in \mathbb{R}$. The calibration dataset spans the concentration range of interest, with denser sampling in the pale-yellow region.

6 Gaussian Process Regression

6.1 Model

A Gaussian Process prior is placed over the function $c = f(\mathbf{f})$:

$$f \sim \mathcal{GP}\big(\mu(\mathbf{f}),\; k(\mathbf{f}, \mathbf{f}')\big)\,. \tag{13}$$

The mean function μ is absorbed into a normalisation of the target (zero-mean after standardisation). The kernel is a product of two anisotropic squared-exponential (RBF) kernels — one for the distance features, one for the temperature features — plus a white noise term:

$$k(\mathbf{f}, \mathbf{f}') = \sigma_f^2\; k_{\text{dist}}(\mathbf{f}_d, \mathbf{f}'_d)\; k_{\text{temp}}(\mathbf{f}_T, \mathbf{f}'_T) + \sigma_n^2\, \delta(\mathbf{f}, \mathbf{f}'), \tag{14}$$

$$k_{\text{dist}}(\mathbf{f}_d, \mathbf{f}'_d) = \exp\left(-\frac{1}{2}\sum_{ij}\frac{(d_{ij} - d'_{ij})^2}{\ell_{ij}^2}\right), \tag{15}$$

$$k_{\text{temp}}(\mathbf{f}_T, \mathbf{f}'_T) = \exp\left(-\frac{1}{2}\sum_{i}\frac{(T_i - T'_i)^2}{\ell_{T,i}^2}\right). \tag{16}$$

Here $\mathbf{f}_d$ and $\mathbf{f}_T$ are the distance and temperature sub-vectors of $\mathbf{f}$; σ_f^2 is the signal variance; σ_n^2 is the observation noise variance; and ℓ_{ij}, $\ell_{T,i}$ are per-feature length scales. All hyperparameters are optimised by maximising the log marginal likelihood:

$$\log p(\mathbf{c} \mid \mathbf{F}) = -\frac{1}{2}\mathbf{c}^\top\big(\mathbf{K} + \sigma_n^2\mathbf{I}\big)^{-1}\mathbf{c} - \frac{1}{2}\log\det\big(\mathbf{K} + \sigma_n^2\mathbf{I}\big) - \frac{M}{2}\log 2\pi, \tag{17}$$

where $\mathbf{K}_{mm'} = k(\mathbf{f}^{(m)}, \mathbf{f}^{(m')})$ is the $M \times M$ kernel matrix and $\mathbf{c} = (c^{(1)}, \ldots, c^{(M)})^\top$.

6.2 Prediction

Given a new feature vector $\mathbf{f}^*$, the GP posterior predictive distribution is Gaussian with mean and variance

$$\hat{c} = \mathbf{k}_*^\top \left(\mathbf{K} + \sigma_n^2 \mathbf{I}\right)^{-1} \mathbf{c}, \tag{18}$$

$$\hat{\sigma}^2 = k(\mathbf{f}^*, \mathbf{f}^*) - \mathbf{k}_*^\top \left(\mathbf{K} + \sigma_n^2 \mathbf{I}\right)^{-1} \mathbf{k}_*, \tag{19}$$

where $(\mathbf{k}_*)_m = k(\mathbf{f}^*, \mathbf{f}^{(m)})$. The estimate $\hat{c}$ is the predicted concentration and $\hat{\sigma}$ is the corresponding 1σ uncertainty.

A measurement is flagged as potentially extrapolated if $\mathbf{f}^*$ lies outside the per-feature range of the calibration set $\mathcal{D}$.

7 PCA Projection (Supplementary)

When the colour gamut varies along essentially one physical axis (white – yellow – brown driven by a single chemical), the between-reference covariance has one dominant eigenvalue and a 1-D projection suffices for a scalar calibration curve.

Let the N_c normalised reference-patch colour vectors $\{\bar{\mathbf{x}}_r\}_{r \in \mathcal{R}}$ have centroid $\bar{\mathbf{x}} = \frac{1}{|\mathcal{R}|} \sum_r \bar{\mathbf{x}}_r$. The between-reference covariance is

$$\mathbf{C} = \frac{1}{|\mathcal{R}| - 1} \sum_{r \in \mathcal{R}} (\bar{\mathbf{x}}_r - \bar{\mathbf{x}})(\bar{\mathbf{x}}_r - \bar{\mathbf{x}})^\top. \tag{20}$$

Its eigendecomposition $\mathbf{C}\mathbf{v}_k = \lambda_k \mathbf{v}_k$ yields eigenvectors sorted in descending order. The scalar projection of a colour vector onto the principal axis is

$$s = \mathbf{v}_1^\top (\mathbf{x} - \bar{\mathbf{x}}). \tag{21}$$

The explained-variance ratio $\lambda_1 / \sum_k \lambda_k$ quantifies how well the 1-D approximation holds. If this ratio is close to 1, the GP distance-based approach and the PCA scalar approach will give nearly identical results; the GP approach is more general and is preferred.

8 Summary of Measurement Procedure

1. For each available LED i, capture a Bayer image of the test strip on the stage.
2. For each responsive channel $k \in \mathcal{C}_i$, extract the patch means for SAMPLE and all reference patches.
3. Optionally update the calibration covariance using equation (8) with the live reference pixels.
4. Compute the normalised colour vectors (equation (2)) for the SAMPLE and each reference patch.
5. Compute the Mahalanobis distances d_{ij} (equation (9)) and check for outliers (equation (10)).
6. Read the LED temperature T_i for each LED.

7. Assemble the feature vector $\mathbf{f}$ (equation (11)).
8. Query the fitted GP to obtain $(\hat{c}, \hat{\sigma})$ (equations (18) and (19)).
9. Report $\hat{c}$ as the concentration estimate with $\hat{\sigma}$ as the 1σ uncertainty. Raise warnings for chi-squared outliers or extrapolation.

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